

Name _____

Centre
Number

Candidate
Number

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ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2007

BIOLOGY
HIGHER 1
PAPER 1

8875 / 1
28 AUG 2008

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number on the multiple choice answer sheet provided.

There are **thirty** questions in this paper. Answer **all** questions. For each question there are four possible answers, **A, B, C** and **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

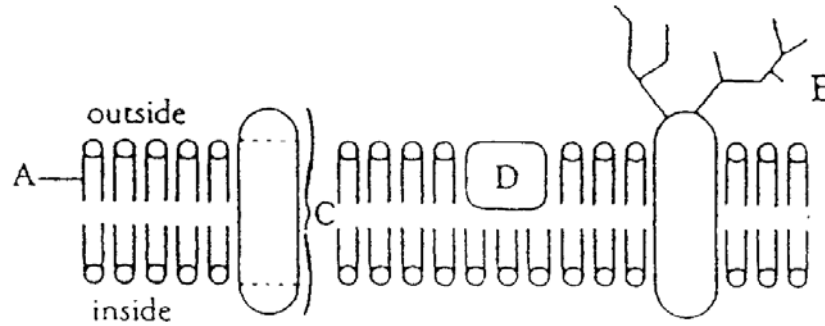
Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

This question paper consists of 11 printed pages

For Q1-2, refer to the following diagram which shows part of a cell membrane.



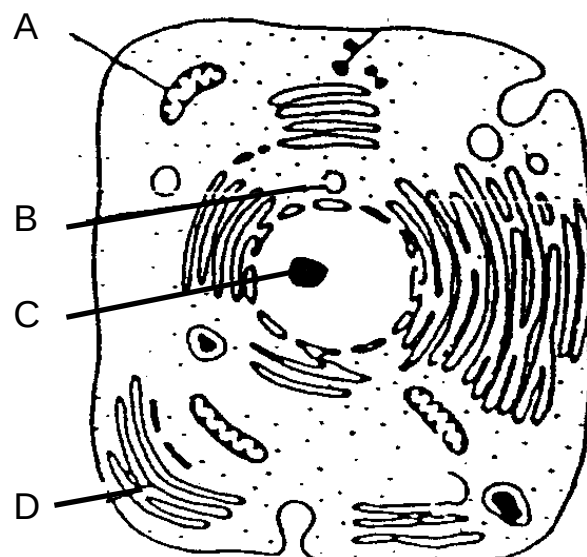
1. Which structure(s) contain(s) the presence of a peptide bond?

- A** A only
- B** B only
- C** A and B
- D** C and D

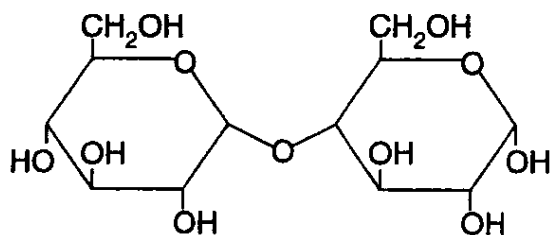
2. Which structure(s) restricts the entry of hydrophilic substances into the cell?

- A** **A only**
- B** B only
- C** A and C
- D** B and C

3. The diagram below shows an animal cell. Which of the following is responsible for the synthesis of rRNA?

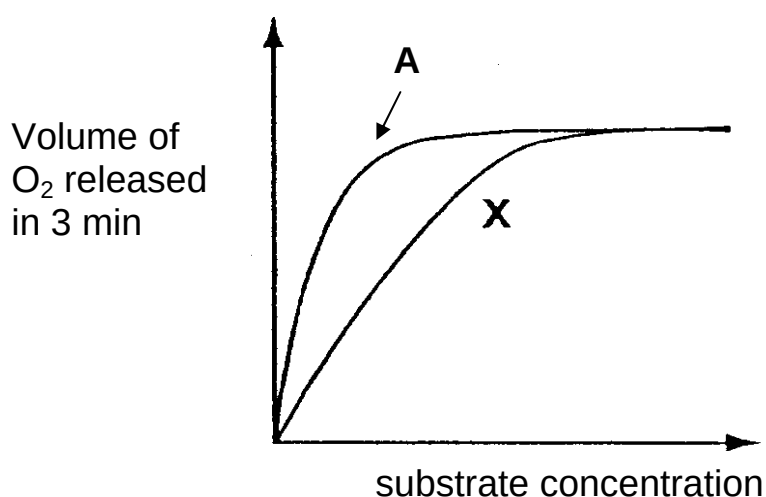


4. The diagram below shows a disaccharide with a particular type of glycosidic bond.



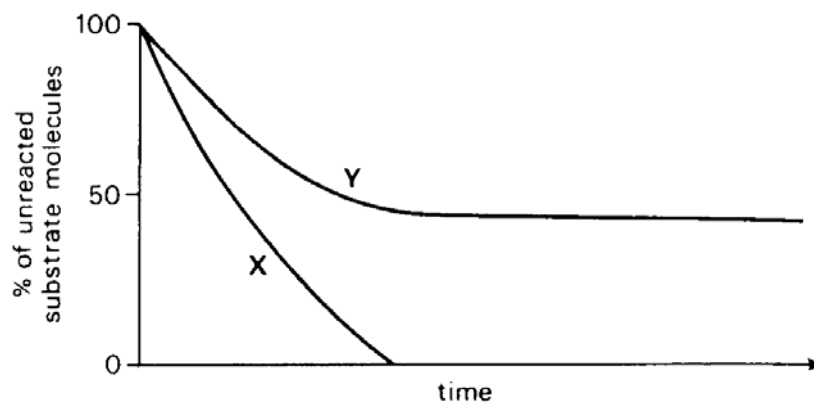
Which of the following molecules contain this type of bond?

- A amylose only
 - B amylose and cellulose
 - C cellulose only
 - D neither amylose nor cellulose
5. Catalase was added to 15ml of hydrogen peroxide at 37°C, and the results were plotted on Graph A. Which of the following would result in graph X?



- A Addition of a competitive inhibitor
- B Addition of a non-competitive inhibitor
- C Addition of less catalase
- D Heating to 48°C

6. Graph X shows the results of an enzyme reaction.



Which of the following would result in Graph Y?

- A Addition of a non-competitive inhibitor
 - B Decreased substrate concentration
 - C **End-product inhibition**
 - D Increased substrate concentration
7. Which of the following catalyses the formation of a peptide bond during translation?
- A Aminoacyl-tRNA synthetase
 - B DNA polymerase
 - C **Peptidyl transferase**
 - D RNA polymerase
8. The table below shows some amino acids and their DNA genetic code.

Amino acid	DNA code
Alanine	CGG
Lysine	TTT
Arginine	GCG
Phenylalanine	AAA
Glycine	CCA
Valine	CAA

A DNA sequence codes for a polypeptide arginine-glycine-phenylalanine-valine-alanine.

The following mRNA molecule was transcribed from the DNA sequence above. Which mRNA triplet contains a single transcription error?

CGCGUUUUUGUUGCC

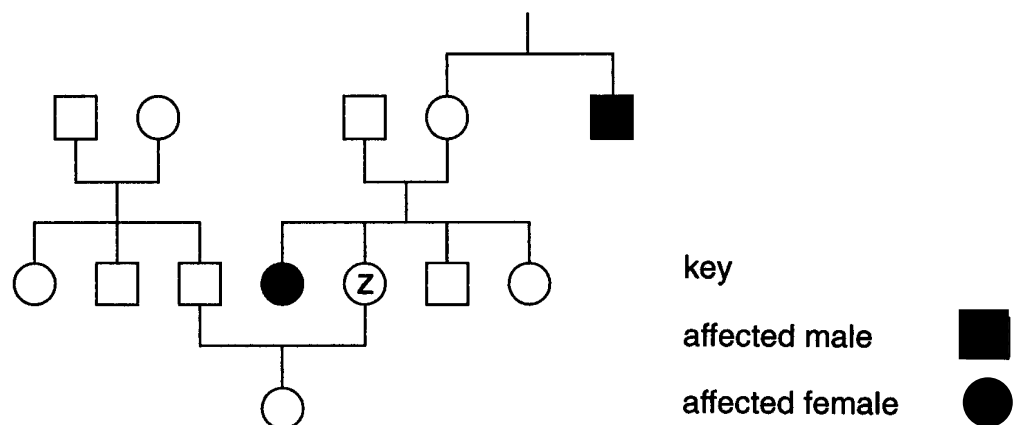
- A The first
- B **The second**
- C The third
- D The fourth

9. How does DNA synthesis along the lagging strand differ from that on the leading strand?
- A An RNA primer is needed on the lagging strand, but not on the leading strand.
 - B Deoxyribonucleotides are added to the 5' end instead of the 3' end.
 - C Helicase synthesizes Okazaki fragments, which are then linked together.
 - D **Okazaki fragments, synthesized 5' → 3', are linked by DNA ligase.**

10. Red-green colour-blindness is a sex-linked, recessive trait. A colour-blind man marries a woman with normal vision, whose father is colour-blind. If they have a daughter, what is the probability that she will be colour-blind?

- A 0
- B $\frac{1}{4}$
- C **$\frac{1}{2}$**
- D $\frac{3}{4}$

11. Galactosaemia is a condition caused by a recessive allele of an autosomal gene in which galactose cannot be metabolised. The diagram shows the pattern of inheritance of galactosaemia in a family.



What is the chance that individual Z is a carrier of the condition?

- A $\frac{1}{2}$
- B $\frac{1}{3}$
- C $\frac{1}{4}$
- D **$\frac{2}{3}$**

12. A plant is heterozygous for a pair of alleles that are codominant. This plant is self-pollinated and the resulting seeds are germinated and allowed to grow. Which ratios are expected in the offspring?

	ratio of phenotypes	ratio of genotypes
A	1 : 2 : 1	1 : 2 : 1
B	1 : 2 : 1	3 : 1
C	3 : 1	1 : 2 : 1
D	3 : 1	3 : 1

Ans: A

13. A group of small fish live in a lake with a uniformly light-brown sandy bottom. Most of the fish are light brown, but about 10% are mottled. This fish species is often prey for large birds that live on the shore. A construction company dumps a load of gravel in the bottom of the lake, giving it a mottled appearance. Which of these statements presents the most accurate prediction of what will happen to this fish population?

- A** As the mottled fish are eaten, more will be produced to fill the gap.
- B** In two generations, all the fish will be mottled.
- C** **The proportion of mottled fish will increase over time.**
- D** The ratio of 9 light brown: 1 mottled fish will not change.

14. Which of the following pairs is *least* likely to represent homology?

- A** The haemoglobin of a baboon and that of a gorilla
- B** The mitochondria of a plant and those of an animal
- C** **The wings of a bird and those of an insect**
- D** The wings of a bat and the arms of a human

15. Which one of the following structures would be found in an animal cell undergoing mitosis, but not in a plant cell undergoing mitosis

- A** **centriole**
- B** centromere
- C** centrosome
- D** chromosome

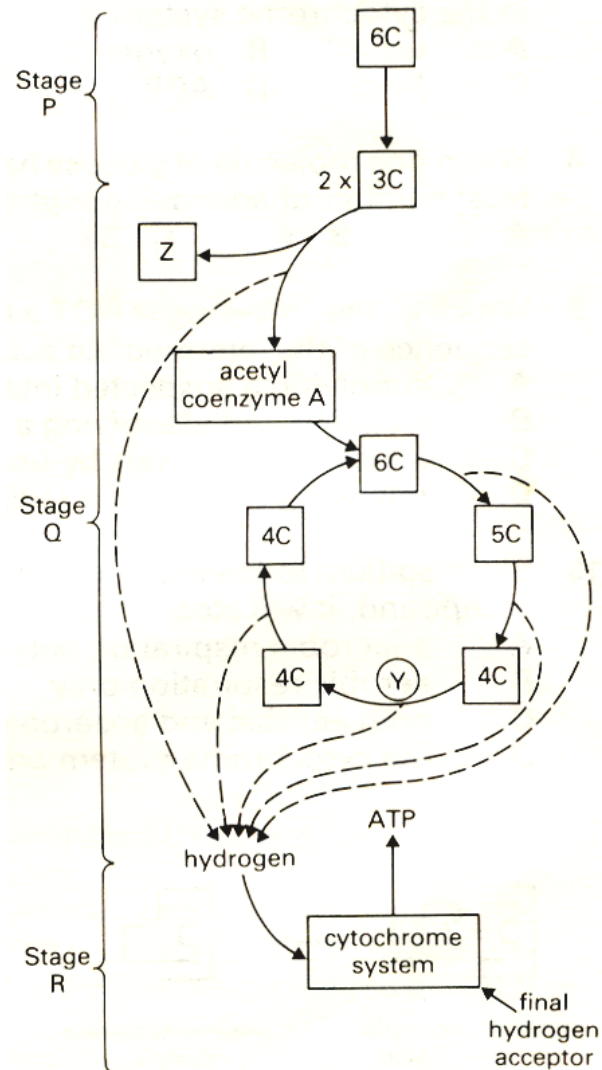
16. During which phase in somatic cell division does replication of chromosomes take place?

- A** early prophase
- B** **interphase**
- C** late prophase
- D** metaphase

17. How many moles of carbon dioxide are produced by the complete oxidation of 1 mole of pyruvate?

- A 1
- B 2
- C 3**
- D 6

18.



Which of the following indicates the correct locations at which stages **P**, **Q** and **R** occur in a cell?

	Cristae	Cytosol	Matrix
A	R	P	Q
B	Q	P	R
C	P	R	Q
D	R	Q	P

19. Some of the photosynthetic organisms containing chloroplasts that lack PSII (photosystem II) are able to survive. The best way to detect the lack of PSII in these organisms would be
- A Determine if they have thylakoids in the chloroplasts
 - B Test for liberation of oxygen in the light**
 - C Test for carbon fixation in the dark
 - D Determine the production of starch
20. Which of the following processes could still occur in a chloroplast in the presence of an inhibitor that prevents H^+ from passing through ATP synthase complexes?
- 1 Sugar synthesis
 - 2 Photolysis of water
 - 3 Transfer of electrons down the electron transport chain
 - 4 Oxidation of NADPH
- A 1 and 2
 - B 1 and 4
 - C 2 and 3**
 - D 3 and 4
21. Which of the following statements about chromosome structure is true?
- 1 Linker DNA refers to the linear double-stranded DNA between adjacent nucleosomes.
 - 2 The 30 nm chromatin fibre has a solenoid structure.
 - 3 The 300 nm chromatin fibre is attached to multiple locations on a central protein scaffold.
 - 4 Euchromatin is the more diffuse region of the interphase chromosome and is transcriptionally inactive.
- A 1 and 2**
 - B 1 and 4
 - C 2 and 3
 - D 3 and 4
22. Which of the following statements is true of the end-replication problem of linear DNA?
- 1 When a linear DNA molecule replicates, a gap is left at the 3' end of each new strand because DNA polymerase can only add nucleotides to a 5' end.
 - 2 Telomerase prevents the end-replication problem from occurring.
 - 3 Repeated rounds of replication produce shorter and shorter DNA molecules.
 - 4 Prokaryotes do not have the end-replication problem.
- A 1 and 2
 - B 1 and 4
 - C 2 and 3
 - D 3 and 4**

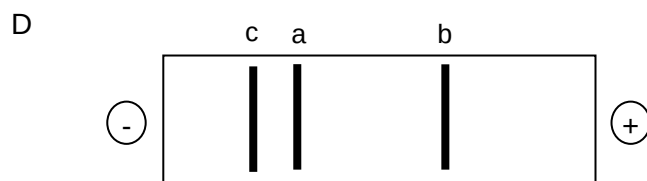
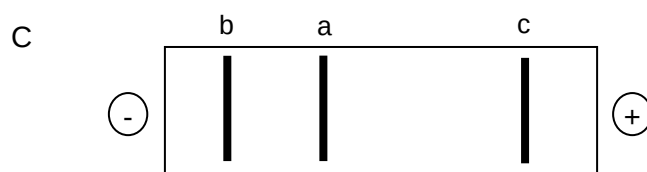
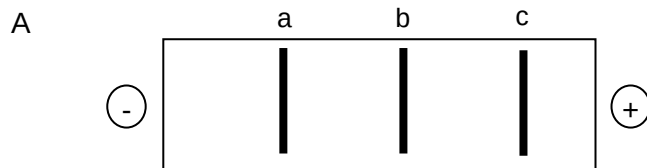
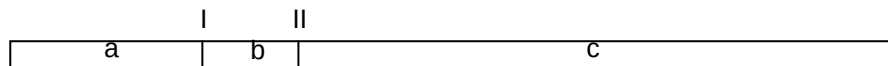
23. A virus has a base ratio of $(A + G) / (U + C) = 1$. What type of virus is this?

- A a double-stranded DNA virus
- B a double-stranded RNA virus**
- C a single-stranded DNA virus
- D a single-stranded RNA virus

24. Which of the following is a requirement for the mechanism of RNA splicing?

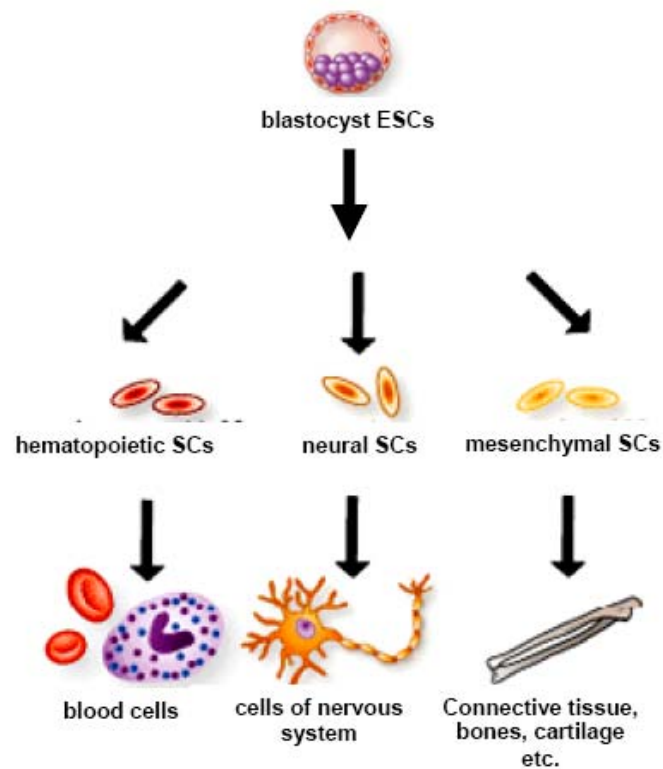
- A RNA polymerase
- B 3' poly-adenylation
- C splice site**
- D 5' modified guanosine cap

25. This segment of DNA has restriction sites I and II, which create restriction fragments a, b, and c. Which of the following gels produced by electrophoresis would represent the separation and identity of these fragments?



26. A plasmid has two antibiotic resistance genes, one for ampicillin and one for tetracycline. It is treated with a restriction enzyme that cuts in the middle of the ampicillin gene. DNA fragments containing a human globin gene were cut with the same enzyme. The plasmids and fragments are mixed, treated with ligase, and used to transform bacterial cells. Clones that have taken up the recombinant DNA are the ones that
- A Are blue and can grow on plates with both antibiotics.
 - B Can grow on plates with ampicillin but not with tetracycline.
 - C Can grow on plates with tetracycline but not with ampicillin.**
 - D Cannot grow with any antibiotics.
27. The completion of the Human Genome Project revealed that the human genome contains fewer genes than expected, not so many more than simpler organisms. Which of the following statements could explain this phenomenon?
- 1 RNA transcripts of human genes are more likely to undergo alternative splicing.
 - 2 Polypeptide domains are combined in a variety of ways.
 - 3 Post-translational processing adds diversity to the resulting polypeptides.
 - 4 Gene expression patterns in humans are often more complex than those in other organisms.
- A 1 only**
 - B 1 and 2
 - C 2, 3 and 4
 - D 1, 2, 3 and 4
28. To date, genetically modified food crops have been modified to _____.
- A deliver vaccines
 - B produce needed human nutrients
 - C resist insect pests
 - D all of the above**
29. Which of the following statements regarding stem cells is **false**?
- A One potential side effect of any embryonic stem cell-based therapy is the formation of a tumour.
 - B Stem cells are present within various organs of the adult body.
 - C Stem cells can develop into a whole organism when implanted into the womb.**
 - D Stem cells can be grown and expanded indefinitely in culture under appropriate conditions.

30. What feature of embryonic stem cells (ESCs) is illustrated below?



- A ESCs are capable of dividing indefinitely.
- B ESCs are multipotent.
- C ESCs are pluripotent.**
- D ESCs are totipotent.